

Bug Report

Defects raised from the Playwright regression cycle of 03 October 2026. Eight entries: one reproduced with screenshot, video and trace; seven raised from observed behaviour during the automation build.

1. Document control

Document	Bug Report — PMCH Healthcare ERP Test Automation
Version	1.0
Date raised	03 October 2026
Raised by	QA Automation — EMSL Dev Team
Test cycle	Regression run of 03 Oct 2026, 08:04 UTC — 24 tests in 3 m 18 s, 23 passed, 1 failed.
Companion document	docs/TEST_CASES.pdf — the 35 cases these defects were found against.
Evidence	docs/assets/failures/ — screenshots, videos, Playwright traces and page snapshots, with manifest.json describing the run. Regenerate with <code>npm run docs:evidence</code> .
Video playback	This PDF cannot play video. Open docs/src/bug-report.html in a browser for the inline players, or open the .webm files directly — paths are given with each defect and in section 5.

2. Environment

Application	HIS ERP — Healthcare Information System (Blazor Server / MudBlazor)
Tenant	Popular Medical College and Hospital Limited
Host	Hospital server over HTTPS, port 9006 (BASE_URL in .env). Live database.
Account	Administrator (credentials held in .env, not in this document)
Client	Chromium (Desktop Chrome profile), 1920×1080, headed, single worker
Harness	Playwright Test 1.63 · TypeScript · Windows 11

3. Legend

Severity	Critical — data loss or the module cannot be used · High — a core workflow produces a wrong result · Medium — workflow completes but behaves wrongly or needs a workaround · Low — cosmetic.
Priority	P1 fix before release · P2 fix this cycle · P3 backlog.
Evidence strength	REPRODUCED captured in the 03 Oct run with artifacts attached · OBSERVED seen repeatedly while building the suite and recorded in the test code, but not captured as an artifact in this run.

4. Defect summary

ID	TITLE	MODULE	SEVERITY	PRI	EVIDENCE
BUG-001	Patient Registration list opens empty — Show From / Show To default to today	Registration	Medium	P2	REPRODUCED screenshot, 2 videos, trace
BUG-002	Advance payment taken in the admission wizard never reaches the Advance Dashboard	Admission / Accounts	High	P1	OBSERVED across runs
BUG-003	Brute-force throttle answers <i>every</i> request from the IP with HTTP 429	Authentication	Medium	P2	OBSERVED , measured
BUG-004	Patient Registration grid can take over 30 seconds to render its first row	Registration	Medium	P2	REPRODUCED run timings
BUG-005	Blazor circuit is dropped when more than one browser session is open	Platform	Medium	P2	OBSERVED
BUG-006	Every module route paints "Access Denied" for a third of a second before the real page	Platform	Medium	P2	OBSERVED
BUG-007	ERP host is served over HTTPS with a self-signed certificate	Infrastructure	Medium	P2	OBSERVED
BUG-008	Two misspelled buttons on the create-patient wizard: "Registration", "Previews"	Registration	Low	P3	OBSERVED

One further item is an **automation defect**, not an application defect, and is tracked separately in section 6: TC_ADM_002 assumes the registration grid is never empty, which is why BUG-001 took the run down with it.

5. Defect detail

BUG-001 — Patient Registration list opens empty because the date filter defaults to today

Module / page	Registration > Patient Registration (/hospital/newregistration)
Severity / Priority	Medium / P2
Status	OPEN — reproduced on both attempts of the 03 Oct run
Found by	TC_ADM_002 — tests/regression/admission.regression.spec.ts:52
Reproducibility	Deterministic. Any day before the first patient of the day is registered — in this run, both attempts, 08:04 and 08:04 UTC.

STEPS TO REPRODUCE

1. Sign in as Administrator.
2. Click the **REGISTRATION** tile on the module launcher.
3. Click **PATIENT REGISTRATION** in the left drawer.
4. Look at the grid without touching the date filter.

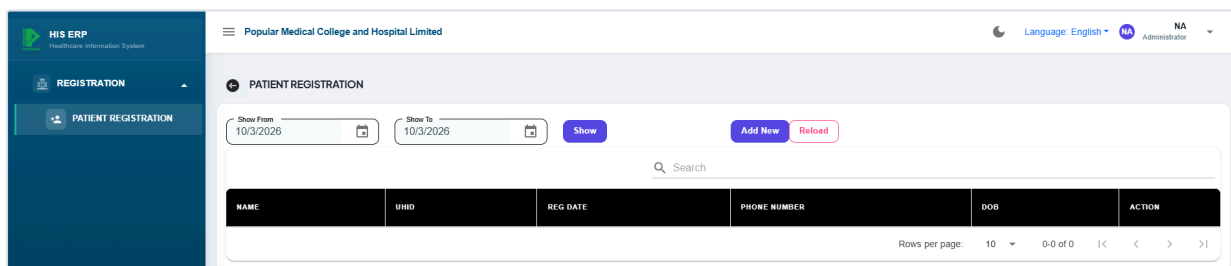
EXPECTED RESULT

The patient list opens on a range that shows work in progress — the recent registrations — so a user arriving at the page can find and open a patient record without first reaching for the date pickers.

ACTUAL RESULT

Show From and **Show To** are both set to today (10/3/2026), so the grid shows only patients registered today. Before the day's first registration that is nobody: the headers render, the body is empty, and the pager reads **0-0 of 0**. The page gives no indication that a filter is responsible — there is no empty-state message and no hint that widening the range would bring rows back.

EVIDENCE



Screenshot — assets/failures/TC_ADM_002/screenshot-1.png, cropped to the filter bar and the grid; the file itself is the full page. Both date filters read **10/3/2026**, the grid body is empty under its six headers, and the pager reads **0-0 of 0**.

Video, attempt 1 — assets/failures/TC_ADM_002/video-1.webm (479 KB). The page loads, the grid paints its headers, and nothing arrives for the 30 seconds the test waits. Not playable in the PDF — open the HTML or the file itself.

Video, attempt 2 (retry) — assets/failures/TC_ADM_002/video-2.webm (525 KB). Identical behaviour on the retry, which is what rules out a transient load failure.

- **Playwright trace** — assets/failures/TC_ADM_002/trace-2.zip. Open with `npx playwright show-trace docs/assets/failures/TC_ADM_002/trace-2.zip` for the DOM snapshot, network log and console at each step.
- **Page snapshot** — assets/failures/TC_ADM_002/error-context-1.md and error-context-2.md. The accessibility tree at the moment of failure, including the empty row groups and the **0-0 of 0** pager.

ERROR AS THE RUN REPORTED IT

```
TimeoutError: locator.innerText: Timeout 30000ms exceeded.  
Call log:  
  - waiting for locator('table.mud-table-root')  
    .locator('tbody tr').first().locator('td').nth(1)  
  
at tests/regression/admission.regression.spec.ts:32  
  const uhid = (await registrationPage.rows.first()  
    .locator('td').nth(1).innerText()).trim();
```

ANALYSIS

The grid did not fail to load — it loaded correctly and returned nothing, because the filter it loaded with asks for one day. The run itself proves the filter is the cause rather than the data: TC_REG_004, which asserts the same grid has at least one row, *passed* in the same run. It runs after TC_FLOW_001, which registers a patient — so by the time TC_REG_004 looked, there was one registration dated today and the same default filter returned it. TC_ADM_002 ran before that, and saw an empty day.

SUGGESTED FIX

- Default **Show From** to a range that covers recent work — the last 7 or 30 days — rather than today, with **Show To** staying at today.
- Failing that, render an empty state in the grid that names the filter: *"No patients registered between 10/3/2026 and 10/3/2026. Widen the date range to see earlier registrations."*

BUG-002 — Advance payment taken in the admission wizard never reaches the Advance Dashboard

Module / page	Hospital > New Admission (/hospital/patientadmission) → Hospital > Advance Dashboard
Severity / Priority	High / P1 — money is collected at the counter and does not appear against the admission
Status	OPEN — needs confirmation from the module owners
Found by	TC_FLOW_001 — tests/regression/admission.regression.spec.ts:121
Reproducibility	Every run of the end-to-end flow so far.

STEPS TO REPRODUCE

1. Sign in as Administrator and register a patient.
2. Open **New Admission** and search the new UHID.
3. Fill the **Contact Person** tab.
4. On **Advance Payment**, enter all three: card VISA / EBL POS **1000**, bKash **2000**, cash **3000**.
5. Confirm the **Total:** panel reads **6000**.
6. Complete **Admission Detail** and press **Create Patient Admission**.
7. Open **Hospital > Advance Dashboard** and look for the admission.

EXPECTED RESULT

The 6000 taken at admission is recorded against the new admission and shows on the Advance Dashboard, so the billing desk can see what the patient has already paid.

ACTUAL RESULT

The admission is created correctly — it appears on **Patient Dashboard For Billing Dept** with the right bed, ward, mobile and blood group — but the Advance Dashboard carries **no row** for it. The wizard's own Total panel had summed the three amounts to 6000 immediately before the submit, so the figures were accepted by the form and lost somewhere between the submit and the advance ledger.

EVIDENCE

- No screenshot: the run that produced this observation predates the evidence collector, and TC_FLOW_001 passes, so Playwright captures nothing for a passing test.
- The observation is recorded in the test itself, at tests/regression/admission.regression.spec.ts:189, as the reason the advance total is deliberately not asserted after the submit.
- The pre-submit total is asserted (Total: 6000), so every passing run is standing evidence that the amounts reached the form.

OPEN QUESTION FOR THE MODULE OWNERS

It is not yet established whether this is a defect or a missing step: **Advance Collection** is its own page, and it may be that the wizard's Advance Payment tab only records an intent and the money has to be receipted separately. The answer decides whether this is a code fix or a documentation and training issue — and either way the wizard should not present a Total that then goes nowhere.

SUGGESTED FIX

- If the wizard is meant to post the advance: fix the posting and add the Advance Dashboard assertion to TC_FLOW_001 so the fix stays fixed.

- If it is not: remove or relabel the Advance Payment tab so it does not read as money taken, and state where the receipt is actually raised.

BUG-003 — Brute-force throttle answers every request from the IP with HTTP 429, not just login attempts

Module / page	Authentication — affects the whole application
Severity / Priority	Medium / P2
Status	OPEN
Found by	The negative login cases (TC_LOGIN_003-007), whose after-effects reached unrelated pages.
Reproducibility	Consistent, after roughly seven failed logins from one address.

STEPS TO REPRODUCE

1. From one IP address, attempt to log in with a wrong password about seven times.
2. From the *same* IP, on a different machine or browser, request any page — including one an already-signed-in user is working on.

EXPECTED RESULT

Throttling protects the sign-in endpoint and the account being attacked. Other users on the same network, and other routes, keep working.

ACTUAL RESULT

The host answers **every** request from that address with **HTTP 429** and `Retry-After: 60`. Measured, it clears in about ten seconds rather than the sixty advertised. During the cool-off a Blazor page load does not degrade gracefully: the circuit never starts, so the page renders as dead chrome rather than showing a "too many attempts, try again shortly" message.

WHY IT MATTERS BEYOND THE TEST SUITE

A hospital floor is behind one NAT address. One person mistyping their password several times locks out *everyone* on that floor for the cool-off — reception, billing and the wards alike — with no message explaining why the ERP has stopped responding.

EVIDENCE

- Worked around in `utils/helpers.ts` — `gotoTolerant()` and `reloadTolerant()` retry past the 429, and their comments record the measured behaviour.
- `reloadTolerant()` also documents a second symptom: a plain reload during the cool-off does not surface the 429 at all, it throws `net:ERR_HTTP_RESPONSE_CODE_FAILURE`.

SUGGESTED FIX

- Scope the throttle to the sign-in endpoint and the account being attempted, not to every route from the IP.
- Make `Retry-After` truthful — advertise the ten seconds actually enforced.
- Serve a readable 429 page instead of letting the Blazor circuit fail silently.

BUG-004 — Patient Registration grid can take over 30 seconds to render its first row

Module / page	Registration > Patient Registration (/hospital/newregistration)
Severity / Priority	Medium / P2
Status	OPEN — measured in the 03 Oct run
Found by	TC_REG_004 — tests/regression/registration.regression.spec.ts:71

STEPS TO REPRODUCE

1. Sign in as Administrator.
2. Open **Registration > Patient Registration**.
3. Time how long it takes for the first row to appear, on a range that has data.

EXPECTED RESULT

A single page of ten rows renders in a couple of seconds, like every other page in the suite.

ACTUAL RESULT

TC_REG_004 took **33.8 seconds** in the 03 Oct run, against a suite median of about 2 seconds. It does nothing but open the page, read six column headers and wait for one row — the time is the grid. The same slowness is why the project raises Playwright's default timeouts across the board: 90 s per test, 30 s per action, 45 s per navigation.

EVIDENCE

- Run timings in docs/assets/failures/manifest.json, and in the HTML report under playwright-report/. Comparison, same run, same session: TC_REG_003 (same page, no row wait) 2.4 s; TC_REG_002 4.4 s; TC_REG_008 3.8 s.

SUGGESTED FIX

- Profile the query behind the grid — a 10-row page should not be fetching the whole patient table before paging it.
- Show a loading indicator while the fetch is in flight, so the page does not read as empty while it works.

BUG-005 — Blazor circuit is dropped when more than one browser session is open

Module / page	Platform — the Blazor Server circuit, all modules
Severity / Priority	Medium / P2
Status	OPEN
Found by	Suite configuration — the reason <code>workers: 1</code> is forced in <code>playwright.config.ts</code> .

STEPS TO REPRODUCE

1. Sign in as the same account in two or more browsers at once against this host.
2. Work in both.

EXPECTED RESULT

Each session keeps its own circuit. A second browser does not disturb the first.

ACTUAL RESULT

Sessions are dropped. Running the suite with more than one worker produced lost sessions and failures unrelated to what the tests asserted, which is why the whole suite is pinned to a single worker — and why the Firefox and WebKit projects are commented out rather than run alongside Chromium.

IMPACT

Beyond the suite: a user who opens the ERP on a second machine, or a shared account used at two counters, can knock the other session out. It also puts a ceiling on test throughput — the full suite is serial, so it runs in minutes where it could run in under one.

EVIDENCE

- `playwright.config.ts` — `workers: 1`, with the reason recorded in the comment above it.
- The Firefox and WebKit project blocks in the same file, commented out for the same reason.

SUGGESTED FIX

Review circuit and session handling on the server — in particular whether sessions are keyed by account rather than by circuit, and what the configured circuit retention limits are.

BUG-006 — Every module route paints "Access Denied" before the real page arrives

Module / page	Platform — every module route
Severity / Priority	Medium / P2
Status	OPEN
Found by	Page-object navigation — <code>BasePage.settleAuthorization()</code> in <code>pages/base.page.ts</code> .

STEPS TO REPRODUCE

1. Sign in as Administrator.
2. Navigate to any module route — for example `/hospital/newregistration`.
3. Watch the first half-second of the load.

EXPECTED RESULT

A user with permission never sees a denial. The route renders a loading state, then the page.

ACTUAL RESULT

The app prerenders its **NotAuthorized** template before the Blazor circuit connects, so **"Access Denied"** is on screen within about 50 ms of every module route and is swapped for the real page roughly a third of a second later. The denial is never real — but it is what the user sees first, on every navigation.

IMPACT

Visible to staff as a flash of a permissions error on every page change, which trains people to distrust the screen. It also makes a genuine permission failure indistinguishable from the false one, for a user and for a test: the suite had to be taught to wait for the real page rather than believe the denial, and a run of about one in twenty does come back genuinely denied and needs a second navigation.

EVIDENCE

- `pages/base.page.ts` — `settleAuthorization()`, whose comment records the measured 50 ms / 300 ms timings and the one-in-twenty genuine denial.

SUGGESTED FIX

Render an authorizing/loading state during prerender instead of the `NotAuthorized` template, and only show the denial once authorization has actually been evaluated.

BUG-007 — ERP host is served over HTTPS with a self-signed certificate

Module / page	Infrastructure — the whole host
Severity / Priority	Medium / P2
Status	OPEN
Found by	Suite configuration — <code>ignoreHTTPSErrors: true</code> is required in <code>playwright.config.ts</code> .

STEPS TO REPRODUCE

1. Open the ERP URL in a browser with no certificate exception configured.

EXPECTED RESULT

A certificate the browser trusts. No warning, no exception for the user to click through.

ACTUAL RESULT

The browser presents a certificate warning that the user has to bypass. The test suite has to set `ignoreHTTPSErrors` for the same reason.

IMPACT

Patient data is involved, and the warning is on a page where staff enter credentials. Training people to click through certificate warnings removes the one signal that would tell them about a real interception, and the suite's blanket `ignoreHTTPSErrors` would not catch one either.

SUGGESTED FIX

Issue a certificate from a CA the hospital's machines trust — an internal CA is enough for an internal host. Then drop `ignoreHTTPSErrors` from the config so the suite starts failing if the certificate goes wrong again.

BUG-008 — Two misspelled buttons on the create-patient wizard

Module / page	Registration > Add New — create-patient wizard (/hospital/patients/new)
Severity / Priority	Low / P3
Status	OPEN
Found by	Page object — pages/registration.page.ts:219-222, where both locators match the app's spelling.

STEPS TO REPRODUCE

- 1. Open **Registration > Patient Registration > Add New**.
- 2. Read the wizard footer buttons.

EXPECTED / ACTUAL

EXPECTED LABEL	ACTUAL LABEL	WHERE
Confirm Registration	Confirm Registration	Wizard footer — the primary action, and the most-read label in the module
Previous	Previews	Wizard footer — back navigation between the three tabs

"Previews" is the worse of the two: it is not a misspelling of anything, it is a different word, and it sits next to *Next* where a user is reading quickly.

NOTE FOR WHOEVER FIXES IT

The automation locators currently match both misspellings on purpose, so that they match the live buttons. Correcting the labels will break `pages/registration.page.ts` and `TC_FLOW_001` — please flag the change so the locators are updated in the same release.

6. Automation defect (not an application defect)

AUT-001 — TC_ADM_002 assumes the registration grid is never empty

Where	tests/regression/admission.regression.spec.ts:29-34 (newestUhid())
Severity / Priority	Medium / P2 — it is why the 03 Oct run came back red
Related	BUG-001 (the application behaviour the test runs into)

WHAT IS WRONG

`newestUhid()` opens the patient list and reads the UHID out of the first row without widening the date filter or checking that a row exists. On a day with no registrations yet the grid is legitimately empty (BUG-001), so the read waits the full 30 seconds and the test fails with a timeout that says nothing about what the case was meant to prove. The same unguarded assumption sits in TC_SMK_ADM_003.

WHY IT MATTERS

The failure reads as "the admission wizard is broken" when the wizard was never reached. It also makes the suite's result depend on the time of day it runs, which is the kind of flakiness that gets a red build ignored.

FIX

1. Widen **Show From** to cover the last 30 days before reading the grid, then press **Show** — the same thing a user would do.
2. Assert the grid has at least one row first, with a message that names the real cause: *"no patients in the registration grid for the selected range"*.
3. Keep BUG-001 open regardless. The test working around the default filter does not make the default filter right.

7. Evidence index

Everything below is under `docs/assets/failures/` and was written by `npm run docs:evidence`, which reads the Playwright HTML report from the last run. Re-running it rebuilds the folder for whatever the latest run contains.

FILE	SIZE	WHAT IT SHOWS
TC_ADM_002/screenshot-1.png	52 KB	The empty grid at the moment of failure, both date filters on 10/3/2026, pager 0-0 of 0.
TC_ADM_002/screenshot-2.png	52 KB	The retry. Byte-identical to attempt 1 — the behaviour is deterministic.
TC_ADM_002/video-1.webm	479 KB	Attempt 1, full recording: navigation, grid paint, the 30 s wait.
TC_ADM_002/video-2.webm	525 KB	Attempt 2 (retry), same sequence.
TC_ADM_002/trace-2.zip	1.7 MB	Playwright trace of the retry — DOM snapshots, network, console, per-step timings. <code>npx playwright show-trace <file></code>
TC_ADM_002/error-context-1.md	12 KB	Accessibility-tree snapshot of the page at failure, plus the failing source lines.

FILE	SIZE	WHAT IT SHOWS
TC_ADM_002/error-context-2.md	12 KB	The same for the retry.
manifest.json	3 KB	Run statistics, per-attempt status and timings, and the error text, as machine-readable JSON.

8. Recommended order of work

ORDER	ID	REASON
1	BUG-002	Money is collected and not recorded. Highest value, and it needs a decision from the module owners before any code is written.
2	BUG-001 + AUT-001	One day's work between them, and together they turn the suite green and make the patient list usable first thing in the morning.
3	BUG-003	One mistyped password can stall a whole floor behind the same NAT address.
4	BUG-006, BUG-004	Both are visible to staff on every working day — a false permissions error on every navigation, and a grid that takes half a minute.
5	BUG-007, BUG-005	Infrastructure and platform work; schedule with the next maintenance window.
6	BUG-008	Cosmetic, but they are the two most-read labels in the registration module. Coordinate with the automation locators.

Bug Report v1.0 · Popular Medical College & Hospital Ltd — Healthcare ERP Test Automation · 03 October 2026 · generated from docs/src/bug-report.html by npm run docs.